Higher-order galaxy bias for DESI thin-bin angular clustering

The one-loop bias model of the lssty1xdesi project

November 2026

Abstract

The lssty1xdesi forecast combines LSST Y1 3x2pt with DESI angular galaxy clustering C_ℓ^{gg} in thin redshift bins, computed without the Limber approximation. This note describes the one-loop galaxy bias model of its DESI likelihood and the physics behind each choice. The model is one likelihood option with every term per bin: the real-space one-loop Eulerian bias expansion on a per-bin linear bias, with the quadratic and tidal biases tied to calibrated relations plus group offsets, a k^2 counterterm, the redshift-space counterterm α_2 in the DESI DR1 convention, and two stochastic terms, a white noise SN_0 and a line-of-sight term $\mathrm{SN}_2 k^2\mu^2$. The shipped parameter files carry the fiducial widths ("model B"). A narrower exploration variant, built for $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ with $\alpha_2 = 0$ and half the offset widths ("model A"), is not shipped; it stays in this note as the record of what motivated the fiducial widths. The paper's fiducial model is the fiducial widths with their tilts, $\alpha_2 = 0$ in every bin and every σ_v prior width multiplied by six. Every term projects exactly: α_2 through the Kaiser transfers, SN_2 through the radial derivatives of the bin kernels. That derivative structure anticorrelates the projected SN_2 of neighboring thin bins, a pattern no white noise reproduces; without SN_2, the exploration variant fails on the thin layouts at its own cut. The draft paper's bias table holds the DESI DR1 sampling combinations of Lagrangian biases, not Eulerian biases, so the model takes its fiducials from the calibrations. The loop terms change the galaxy power by up to 7% at the exploration variant's per-bin cut and up to 15% at the shipped mask.

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1 Scope and conventions

The lssty1xdesi project forecasts DESI spectroscopic galaxy clustering (DESI is the Dark Energy Spectroscopic Instrument; DR1, below, its first data release) as angular power spectra C_ℓ^{gg} in thin redshift bins, combined with LSST Y1 3x2pt, the first-year Rubin Observatory combination of cosmic shear, galaxy–galaxy lensing and galaxy clustering. The spectra are computed without the Limber approximation, which ties each multipole to a single wavenumber and fails for thin bins. The DESI spectra are computed on four bin layouts: 76 bins of equal comoving width ($\Delta d_{\rm com} = 62\,$Mpc), and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. The bins fall into six tracer groups: the bright low-redshift galaxies BGS, the luminous red galaxies LRG1–LRG3 and the emission-line galaxies ELG1 and ELG2. A bin of comoving width $\Delta\chi$ resolves radial Fourier modes up to $k_\parallel \simeq \pi/\Delta\chi$, about $0.075\,h\,\mathrm{Mpc}^{-1}$ for the 62 Mpc bins, so the angular analysis recovers most of the three-dimensional information at that scale [10].

Galaxies are biased tracers: on large scales the galaxy overdensity follows the matter overdensity with one amplitude, the linear galaxy bias, $\delta_g = b_1\,\delta$. With linear bias the likelihood models the galaxy power as $P_{gg} = b_1^2 P_{\rm mm}$, with $P_{\rm mm}$ the halofit matter power spectrum (halofit: a fitting formula for the nonlinear matter power, calibrated on N-body simulations), and a linear-bias fit holds only to $k_{\rm max} = 0.05\,h\,\mathrm{Mpc}^{-1}$ on this data vector (`notes/linear_bias_kmax`). This note describes the one-loop model that replaces it: the fiducial parameter values behind the note (Section 2), the physics the model rests on (Sections 3–5), the size of each term (Section 6), the

model, its two width configurations and the paper model (Section 7), the exact projection of each term to thin-bin spectra (Section 8), and the checks (Section 9). The forecasts made with it are in `notes/correlated_bias_prior_v2`, the correlated-prior note: the linear-bias priors and their accuracy test, the fiducial model against $k_{\rm max}$, and where it breaks. The Fisher machinery that scores those forecasts – derivative columns of the data vector, cached once per layout and recombined per model – is the subject of `notes/fisher_machinery`.

Conventions. ℓ is the multipole; k is the comoving wavenumber, in h/Mpc unless a unit is written; z is redshift and $\bar{z}_i$ the mean redshift of bin i; μ is the cosine between the wavevector and the line of sight; $D(z)$ is the linear growth factor, $D(0) = 1$, and $f = d\ln D/d\ln a$ the growth rate, with a the scale factor; $\sigma_8(z)$ is the rms linear density fluctuation in spheres of $8\,h^{-1}$Mpc; Ω_m is the matter density today in units of the critical density; $\bar{n}$ is the tracer number density. The matter overdensity is δ, the galaxy overdensity δ_g, and $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\,\delta$ is the tidal field, with $s^2 = s_{ij}s_{ij}$. A parameter's marginalized one-sigma error is σ (marginalized: quoted after every other parameter has been integrated out). The spectra enter every fit as band powers, averages over multipole bands of width $\Delta\ell = 10$; a band is kept or cut as one unit. $k_{\rm max}$ is quoted at $z = 0$: bin i is cut at $k_{\rm max}/D(\bar{z}_i)$ by the non-Limber UV-fraction criterion [12], which keeps a band while less than 5% of it comes from the power spectrum above that wavenumber. A fit uses the bands that both this cut and the shipped mask keep; the mask stops every bin at $k = 0.1185\,\mathrm{Mpc}^{-1} \simeq 0.176\,h\,\mathrm{Mpc}^{-1}$.

2 Fiducial model and parameter values

This section freezes the fiducial model and the parameter values behind this note, so its record does not depend on configuration files that can change. This note's own figures are theory illustrations: none applies a scale cut, and all are computed in a CCL environment under two setups, A and B below (`SETUP_A` and `SETUP_B` of `plots/make_figure_data.py`), not with the likelihood. Figures 1, 2, 3 and 4 use setup A; Figure 5 and the σ_4 of Eq. (17) use setup B, which matches the likelihood fiducial.

Cosmology. Both setups share $\Omega_m = 0.3$, $\Omega_b = 0.04$, $H_0 = 67.32\,\mathrm{km/s/Mpc}$ and $10^9 A_s = 2.1$, with A_s the amplitude of the primordial curvature power spectrum. Setup A has $n_s = 0.966$ and massless neutrinos, giving $\sigma_8(0) = 0.843$; setup B has the likelihood's $n_s = 0.96605$ and one massive neutrino of 0.06 eV inside Ω_m. Setup A keeps the low-k loop constants (CCL's default); setup B subtracts them (CosmoLike's convention, Section 8.4). The likelihood fiducial, the evaluate override of the project's example yamls, is setup B's cosmology with $w_0 = -1$, $w_a = 0$ and $\tau = 0.0697186$ fixed, and has $\sigma_8(0) = 0.8278$: where this note quotes a setup A value, its σ_8 is the massless-neutrino 0.843, not the likelihood's 0.8278.

Bias. Bin i of group g has $b_{1,i} = B_g/D(\bar{z}_i)$ with $B_g = 1.34$ (BGS), 1.70 (LRG1–LRG3) and 0.84 (ELG1, ELG2); $b_{2,i}$ sits on the Lazeyras calibration, Eq. (7), $b_{s^2,i}$ on co-evolution, Eq. (6), and $b_{\rm 3nl} = 0$; every offset, tilt, counterterm and stochastic amplitude is zero. Both width configurations and the paper model share this fiducial point.

Prior widths as shipped. Table 1 gives the Gaussian widths per group. The tilt boxes of the tilted variants are flat, per unit redshift: ± 100 (Δ_2, Δ_s) and ± 1000 (α_0) in the shipped

files, the paper model's, which fix $\alpha_2 = 0$ in every bin. In the yamls, the fiducial α_0 width is $16.1\,h^{-2}\mathrm{Mpc}^2$ ($0.5/k^2_{\mathrm{max}}$ at the shipped mask's $0.176\,h\,\mathrm{Mpc}^{-1}$); the Fisher notebook instead sets $0.5/k^2_{\mathrm{max}}$ at each cut of its scan, for α_2 too where a fit frees it, and leaves the tilts flat. The SN_2 unit $f_{\mathrm{sat}}\sigma^2_{\mathrm{sat}}$ is $6.084\,(h^{-1}\mathrm{Mpc})^2$ for BGS and LRG and $0.841\,(h^{-1}\mathrm{Mpc})^2$ for ELG.

Fingers-of-God dispersions. The Fingers-of-God smearing of radial positions by virial motions (Section 3.4 defines it) is modeled per bin with a sampled velocity dispersion σ_v. Seven sampled σ_v groups (BGS split at $\bar{z} = 0.255$), with fiducials 371, 358, 357, 348, 343, 304 and 289 km/s (BGS0, BGS1, LRG1–3, ELG1–2) and exploration prior widths 145, 105, 70, 50, 50, 40 and 30 km/s; the paper model multiplies every width by six (870, 630, 420, 300, 300, 240, 180 km/s). The priors are truncated at $\sigma_v = 0$: at the sixfold width the truncation removes 33% of the BGS0 Gaussian and its truncated mean is about 850 km/s, while the Fisher matrices treat every σ_v prior as an untruncated Gaussian of its width about the fiducial.

The covariance fiducial. The Gaussian covariance is built at the likelihood cosmology with massless neutrinos (its one difference), on the bias fiducial above, with the Fingers-of-God smearing at the fiducial σ_v, over 14,000 deg^2; the forecasts rescale its inverse to 12,300 deg^2 (Section 7.5).

Scale cuts. No figure of this note applies a scale cut. The forecast results this note quotes (Section 9) use the UV-fraction cut computed on the paper model's fiducial by the paper notebook, at the final derivative tier (`chains/model_scale_cuts_final.npz`); the linear-bias experiments keep a cut computed on the linear fiducial (`notes/scale_cut_fiducial_dependence` measures the difference between the two).

3 The one-loop galaxy power spectrum

3.1 Real space, Eulerian basis

Perturbation theory expands the matter overdensity in powers of the linear density; a correction built from two powers of the linear power spectrum is a one-loop term, the first order beyond linear theory. At one loop, every rotationally invariant local gravitational observable up to third order in the linear density enters the galaxy density. After renormalization – absorbing the operators' means and their sensitivity to unmodeled small scales into redefined coefficients – the McDonald–Roy basis [1, 2] carries four deterministic bias parameters, a derivative bias and a noise field:

$$\delta_g = b_1\,\delta + \frac{b_2}{2}\left(\delta^2 - \langle\delta^2\rangle\right) + \frac{b_{s^2}}{2}\left(s^2 - \langle s^2\rangle\right) + \frac{b_{\mathrm{3nl}}}{2}\,\psi + b_{\nabla^2}\,\nabla^2\delta + \epsilon, \tag{1}$$

where ψ is the one independent third-order operator that survives at one loop (Appendix A writes it out) and ϵ is the stochastic field, uncorrelated with the long modes. Higher-order bias means the coefficients beyond linear order: b_2 and b_{s^2} at second order and b_{3nl} at third. This is the convention of FAST-PT (the Python code that evaluates the loop integrals as FFT convolutions), of `EulerianPTCalculator` (`pyccl/nl_pt/ept.py`) in CCL, the Core Cosmology Library, and of CosmoLike, the project's compiled C code, which all assemble the auto spectrum

as

$$P_{gg}(k) = b_1^2\,P_{\rm mm}(k) + b_1 b_2\,P_{\delta_1\delta_2}(k) + \tfrac{1}{4}b_2^2\,P_{\delta_2\delta_2}(k) + b_1 b_{s^2}\,P_{\delta_1 s_2}(k) \\ + \tfrac{1}{2}b_2 b_{s^2}\,P_{\delta_2 s_2}(k) + \tfrac{1}{4}b_{s^2}^2\,P_{s_2 s_2}(k) + b_1 b_{\rm 3nl}\,\sigma_3^2(k)\,P_{\rm lin}(k) + b_1 b_{k^2} k^2 P(k), \qquad (2)$$

plus stochastic terms. The five quadratic templates P_{XY} are mode-coupling integrals of two linear spectra over the second-order kernels (Appendix B); $\sigma_3^2(k)$ is the third-order filter of [1, 2]; and b_{k^2} is CCL's name for the derivative bias, degenerate at this order with the effective-field-theory counterterm $\alpha_0 k^2 P_{\rm lin}$. A counterterm is a free coefficient on a fixed k-shape, here $k^2 P_{\rm lin}$, that absorbs what the loop integrals wrongly import from small scales the perturbative model does not describe. The draft paper's expression is exactly the first six terms of Eq. (2), plus $\alpha_0 k^2 P_{\rm lin}$, with $b_{\rm 3nl} = 0$.

Three templates do not vanish at low k: $P_{\delta_2\delta_2} \to 2\sigma_4$, $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$ and $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$, with σ_4 a constant built from the linear spectrum (Appendix B). These constants are white noise, indistinguishable from shot noise. CCL subtracts them only with its `sub_lowk` option (off by default); CosmoLike always subtracts them, and the model is stated in that subtracted convention. The choice matters for a dense sample: at the calibrated biases of BGS at $z = 0.3$ the three constants sum to about $920\,(h^{-1}{\rm Mpc})^3$, five times the Poisson power $1/\bar{n} \simeq 170\,(h^{-1}{\rm Mpc})^3$ of the project's BGS bins there (a sample of discrete objects at number density $\bar{n}$ has the constant power $1/\bar{n}$ even without clustering: the Poisson, or shot-noise, power), so a CCL vector at its default and a CosmoLike vector differ by that white noise. Section 8.4 follows the subtraction to high k, where it leaves a negative constant.

The b_1^2 term carries the full nonlinear matter power (halofit), while the loop templates stay at one-loop standard perturbation theory. Nicola et al. [8] validated this hybrid for an LSST-like projected 3x2pt analysis: with the nonlinear $P_{\rm mm}$ in the b_1^2 term the Eulerian model recovers unbiased cosmology to at least $k_{\rm max} = 0.4\,{\rm Mpc}^{-1}$, while the fully perturbative version fails beyond about $0.2\,{\rm Mpc}^{-1}$. The same work finds significant non-Poisson stochasticity in every galaxy sample it fits (Section 6). The hybrid asks for consistency: the same $P_{\rm mm}$ in the data vector, the covariance and the Fisher derivatives.

3.2 The Lagrangian basis and the exact conversions

An Eulerian expansion like Eq. (1) writes the galaxy field in terms of the evolved matter fields at the time of observation. A Lagrangian expansion writes it instead in terms of the initial linear fields at the positions where the galaxies formed, and lets gravity carry the result forward. The velocileptors code, the DESI DR1 baseline, expands the galaxy field in the initial (Lagrangian) coordinates [4, 5]:

$$F[\delta_0(\mathbf{q})] = 1 + b_1^{\rm L}\,\delta_0 + \frac{b_2^{\rm L}}{2}\left(\delta_0^2 - \langle\delta_0^2\rangle\right) + b_s^{\rm L}\left(s_0^2 - \langle s_0^2\rangle\right) + b_3^{\rm L}\,O_3(\mathbf{q}), \qquad (3)$$

where δ_0 and s_0 are the linear density and tidal fields at the initial time and O_3 is the one independent cubic operator. The quadratic term carries a $1/2$, the tidal term does not. Chen, Vlah and White [4] give the rotation to their Eulerian parameters c_n, defined by $\delta_g = c_1\delta + \frac{c_2}{2}\delta^2 + c_s s^2 + \ldots$ (again no $1/2$ on the tidal term):

$$c_1 = 1 + b_1^{\rm L}, \qquad c_2 = b_2^{\rm L} + \frac{8}{21}\,b_1^{\rm L}, \qquad c_s = b_s^{\rm L} - \frac{2}{7}\,b_1^{\rm L}, \qquad c_3 = b_3^{\rm L} + a\,b_1^{\rm L}, \qquad (4)$$

with a a constant fixed by the choice of third-order operator. Appendix A derives the $8/21$ and $-2/7$ from the second-order co-evolution solution. Mapping the c_n onto the CCL convention of Eq. (1) costs one more step, the factor of two between $c_s s^2$ and $(b_{s^2}/2)s^2$:

$$\boxed{b_1 = 1 + b_1^{\rm L}, \qquad b_2 = b_2^{\rm L} + \frac{8}{21}\, b_1^{\rm L}, \qquad b_{s^2} = 2\, b_s^{\rm L} - \frac{4}{7}\, b_1^{\rm L}, \qquad b_{\rm 3nl} = \frac{32}{315}\, b_1^{\rm L} + 2\, b_3^{\rm L}.} \tag{5}$$

The $b_{\rm 3nl}$ entry uses the third-order co-evolution result of Saito et al. [2] for the dynamically generated part; the $2b_3^{\rm L}$ normalization depends on the choice of O_3 and plays no role here, because every analysis in play sets $b_3^{\rm L} = 0$. A third basis writes the tidal operator as the Galileon $\mathcal{G}_2 = s^2 - \frac{2}{3}\delta^2$ with coefficient $b_{\mathcal{G}_2}$ (written γ_2 by [9]); then $b_{\mathcal{G}_2} = c_s = b_{s^2}/2$ and the quadratic coefficient shifts by $+\frac{4}{3}b_{\mathcal{G}_2}$ (Appendix A).

3.3 Co-evolution and the $b_n(b_1)$ calibrations

If galaxies form with purely local Lagrangian bias ($b_s^{\rm L} = b_3^{\rm L} = 0$) and are then advected by gravity, the Eulerian tidal and third-order biases are generated by the dynamics alone. Setting $b_s^{\rm L} = 0$ in Eq. (5) gives the co-evolution relations [2]

$$b_{s^2} = -\frac{4}{7}\,(b_1 - 1)\,, \qquad b_{\rm 3nl} = \frac{32}{315}\,(b_1 - 1)\,, \tag{6}$$

so every tracer with $b_1 > 1$ has $b_{s^2} < 0$ in this convention. Simulations support the relations at the 10–30% level; the excursion-set calibration used by [9], $b_{\mathcal{G}_2} = 0.524 - 0.547\, b_1 + 0.046\, b_1^2$, gives a $b_{s^2} = 2b_{\mathcal{G}_2}$ about 30% more negative than Eq. (6) at $b_1 \simeq 2.2$. For the quadratic bias the separate-universe fit of Lazeyras et al. [3], valid for $1 \lesssim b_1 \lesssim 10$ and redshift independent, reads

$$b_2^{\rm fit}(b_1) = 0.412 - 2.143\, b_1 + 0.929\, b_1^2 + 0.008\, b_1^3. \tag{7}$$

Figure 1 draws these relations across the six groups at the project fiducial, $b_1 = B_g/D(z)$ with the amplitudes $B_g = 1.34$ (BGS), 1.70 (LRG) and 0.84 (ELG). Each curve spans one group, so its slope shows how far the calibrated biases move across a group: the drift Section 7.2 builds into the model.

3.4 Redshift space

A galaxy's measured redshift carries its peculiar velocity along the line of sight, so the radial position inferred from it is displaced. Positions mapped this way are said to be in redshift space, and the resulting distortions of the clustering are redshift-space distortions. On large scales, coherent infall toward overdensities compresses structures along the line of sight and enhances the radial power: the linear (Kaiser) effect, which multiplies a density mode by $b_1 + f\mu^2$. On small scales, the random virial motions of galaxies inside their halos smear radial positions and damp the radial power: the Fingers of God, which the likelihood models with the sampled dispersions σ_v of Section 2.

In redshift space the one-loop model gains no new bias parameters but several new terms. The DESI DR1 methodology paper [6] writes every full-shape model in the generic form

$$P_{s,g}(k,\mu) = P_{s,g}^{\rm PT}(k,\mu) + \left(\alpha_0 + \alpha_2\mu^2 + \alpha_4\mu^4\right) k^2 P_{\rm lin}(k,\mu) + \left({\rm SN}_0 + {\rm SN}_2\, k^2\mu^2 + {\rm SN}_4\, k^4\mu^4\right). \tag{8}$$

$P_{s,g}^{\rm PT}$ is the one-loop redshift-space spectrum of biased tracers. Its tree-level limit is the Kaiser spectrum $(b_1 + f\mu^2)^2\, P_{\rm lin}$; at one loop every bias operator of Eq. (3) couples to the velocity field,

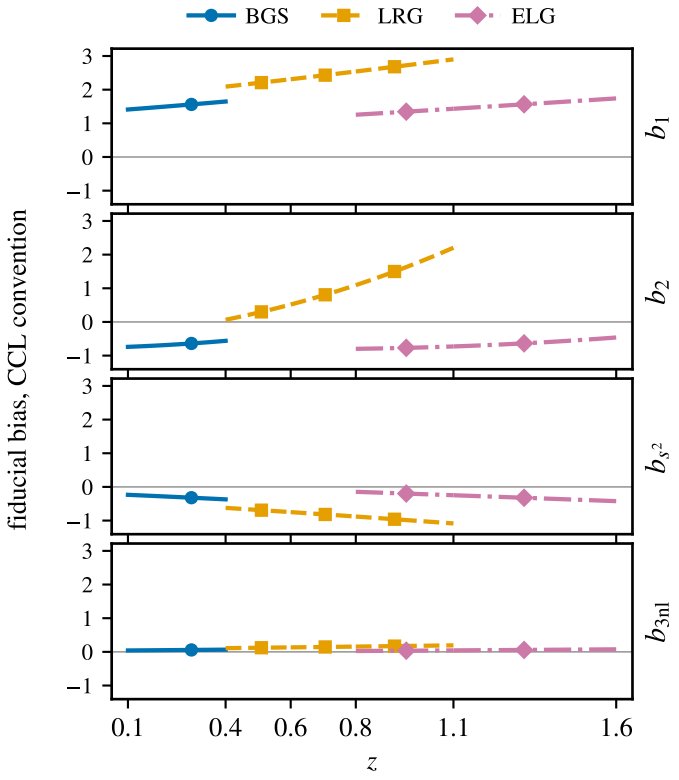


Figure 1: The fiducial biases of the six DESI groups in the CCL convention of Eq. (1), against redshift: the linear bias $b_1 = B_g/D(z)$, the Lazeyras et al. b_2 of Eq. (7), and the co-evolution b_{s^2} and $b_{3\mathrm{nl}}$ of Eq. (6). Each curve spans one group (the redshift ticks are the group edges); the groups of one tracer share B_g, so their curves join. Markers sit at the DESI DR1 effective redshifts; ELG1, which the DR1 full-shape analysis excluded, is marked near its group's center. D is computed at the figure setup A of Section 2 (Appendix C).

producing μ-dependent loop terms with the same four bias coefficients [4, 5]. The counterterms $\alpha_0, \alpha_2, \alpha_4$ absorb the small-scale sensitivity of the loop integrals and the leading velocity dispersion; velocileptors normalizes them as fractional corrections to the linear terms they multiply, so that $\alpha_n = 12.5\,h^{-2}\mathrm{Mpc}^2$ is a 50% correction at $k = 0.2\,h\,\mathrm{Mpc}^{-1}$ [5]. The stochastic terms descend from the noise field: SN_0 is the deviation of the constant noise from the Poisson value, SN_2 the velocity-weighted noise (the leading Fingers-of-God stochastic term) and SN_4 its next moment. A projected analysis keeps the same biases; Section 8 shows which μ-dependent members survive the projection into thin bins.

4 The DESI DR1 full-shape model and priors

The DESI DR1 full-shape analysis [6, 7] fits the power spectrum monopole and quadrupole (the $\ell = 0$ and $\ell = 2$ Legendre moments of $P(k, \mu)$ in μ; the hexadecapole below is $\ell = 4$) of six tracer bins (BGS at $z_{\mathrm{eff}} = 0.295$, LRG1–3 at 0.510, 0.706, 0.919, ELG2 at 1.317, QSO at 1.491) over $0.02 < k < 0.2\,h\,\mathrm{Mpc}^{-1}$. The ELG1 bin ($0.8 < z < 1.1$) failed the fibre-collision systematics

tests and is excluded from the full-shape analysis [6]. The baseline theory is the Eulerian mode of velocileptors, validated against Folps, PyBird and a Gaussian-streaming configuration-space model [6]. The choices that matter here:

- **Basis and sampling.** Even in the Eulerian mode, DESI samples the Lagrangian parameters of Eq. (3), reparameterized as the "physical" combinations

$$\left(1 + b_1^{\rm L}\right) \sigma_8(z_{\rm eff}), \qquad b_2^{\rm L}\, \sigma_8^2(z_{\rm eff}), \qquad b_s^{\rm L}\, \sigma_8^2(z_{\rm eff}), \tag{9}$$

  because the data constrain these products directly and the reparameterization reduces projection effects on σ_8 [6, 7]. $1 + b_1^{\rm L}$ is the Eulerian linear bias, so the first combination is $b_1\sigma_8(z_{\rm eff})$.

- **Priors.** Uniform $[0, 3]$ on $(1 + b_1^{\rm L})\sigma_8$; Gaussian with mean zero and width 5 on $b_2^{\rm L}\sigma_8^2$ and on $b_s^{\rm L}\sigma_8^2$; Gaussian with mean zero and width $12.5\,h^{-2}{\rm Mpc}^2$ on α_0 and α_2 (the 50%-correction-at-$k_{\rm max}$ rule); Gaussian widths of 2 Poisson units on SN_0 and of 5 units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}$ on SN_2, where $f_{\rm sat}$ is the satellite fraction (the share of the sample's galaxies that orbit inside a halo as satellites rather than sitting at its center) and $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units [6, 7].

- **Fixed parameters.** $b_3^{\rm L} = 0$, because the cubic bias is expected to be small for intermediate-mass halos and is degenerate with the counterterms [6, 5]; α_4 and SN_4 drop with the hexadecapole, which DESI DR1 does not fit. The tidal bias is left free rather than tied to co-evolution: tying it would add cosmological information that DESI chose not to claim [6].

- **Budget.** Seven nuisance parameters per tracer $(b_1^{\rm L}, b_2^{\rm L}, b_s^{\rm L}, \alpha_0, \alpha_2, \mathrm{SN}_0, \mathrm{SN}_2)$, 42 in total, with the counterterms and stochastic terms marginalized analytically [7].

Semenaite et al. [9] fit the same DESI DR1 tracers in configuration space jointly with weak lensing, with both tidal biases fixed to calibrated relations in b_1 (the excursion-set $\gamma_2(b_1)$ and the conserved-evolution $\gamma_{21}(b_1, \gamma_2)$) and b_1 and b_2 free per tracer. Their mocks show that fixing the tidal sector to these relations recovers unbiased cosmology for DESI-like tracers: direct evidence for the calibration-anchored tidal bias of the model of Section 7.

5 The draft paper's bias table: the tidal-bias verdict

The draft paper's bias table ("bias parameters per DESI redshift bin") lists, per group, values labeled $b_1^{\rm E}$, $b_2^{\rm E}$, $b_s^{\rm E}$, attributed to the DESI DR1 full-shape analysis. Its $b_s^{\rm E}$ column (positive, 0.6–1.4) contradicts co-evolution, which predicts b_{s^2} between -0.3 and -1.0 for these tracers (Figure 1). The numbers themselves establish where the table comes from.

Three facts identify the table as the DESI physical-basis combinations of Eq. (9) inserted raw:

1. **The linear column divided by $\sigma_8(z_{\rm eff})$ lands on the known biases for four of the five groups.** Figure 2 shows the check: for LRG1–3 and ELG2 the implied Eulerian bias agrees with the project's fiducial $B_g/D(z_{\rm eff})$ to 1–3%. An independent check at a Planck-like cosmology ($\Omega_m = 0.3138$, $\sigma_8 = 0.8076$) finds the same pattern, with ratios of 1.055,

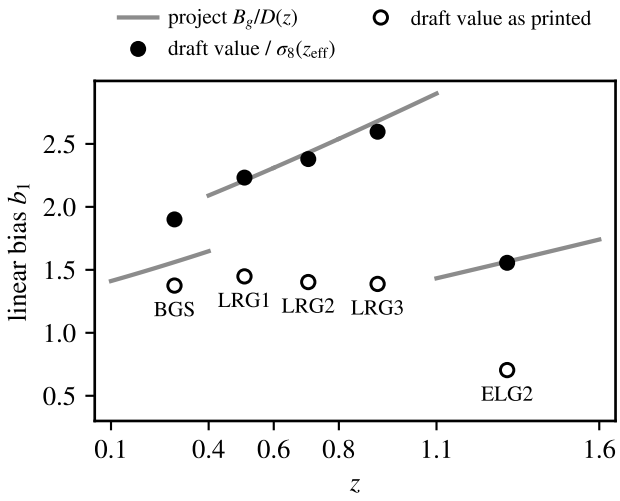


Figure 2: The draft table's linear-bias column ("$b_1^{\rm E}$") read as the DESI physical combination $b_1\sigma_8(z_{\rm eff})$: as printed (open circles) and divided by σ_8 at the group's DESI DR1 effective redshift (filled circles), against the project's fiducial $b_1 = B_g/D(z)$ over each group (gray). $\sigma_8(z)$ is computed at the figure setup A of Section 2 (massless neutrinos, $\sigma_8(0) = 0.843$). The draft has no ELG1 row.

1.022, 1.011 and 1.038. Read directly as Eulerian biases, the column would make the three LRG groups nearly unbiased ($b_1 \simeq 1.4$ where LRGs sit near 2.2–2.7) and ELG2 anti-biased ($b_1 = 0.70$) at $z = 1.3$. The BGS row is the exception: it implies $b_1 \simeq 1.90$ against the project's 1.56, a ratio of 1.22 (1.27 in the Planck-like check). Whether that reflects the project's BGS amplitude $B_g = 1.34$, which comes from the source forecast's configuration rather than a DESI DR1 fit, or a different origin of the row is not established; the BGS entry of the public DESI DR1 chains decides it.

2. **The empty ELG1 row.** DESI DR1 excluded ELG1 from the full-shape analysis (Section 4), so a table transcribed from its products has nothing to put there.

3. **The magnitudes of the other two columns.** Read as $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, the second and third columns (-1.5 to $+0.3$ and $+0.6$ to $+1.4$) sit inside the DESI Gaussian priors of width 5 on exactly those combinations. Read as CCL Eulerian parameters they contradict both co-evolution and the DESI prior centering.

Verdict. The draft's bias table holds the DESI DR1 sampling combinations $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, used as CCL's Eulerian (b_1, b_2, b_{s^2}) without three transformations: removing the σ_8 powers, rotating the Lagrangian parameters to the Eulerian basis with Eq. (5), and the factor of two between the velocileptors and CCL tidal normalizations. The linear column establishes the reading for LRG1–3 and ELG2, and the table's construction extends it to the other two columns. The sign puzzle dissolves: the column was never an Eulerian tidal bias. The cost of the raw reading is large. The tidal term enters with the opposite sign and several times the co-evolution magnitude (the BGS $b_1 b_{s^2}$ product is $+1.96$ against -0.50), the quadratic coefficient differs at a comparable level, and the loop sum moves by several percent of P_{gg} at $k \simeq 0.1\,h\,{\rm Mpc}^{-1}$, the size of the whole correction. The linear column, read as an Eulerian bias,

is 30–55% low for the LRG and ELG groups: a free amplitude absorbs that in a fit, but a data vector, a covariance or a Fisher derivative at a fixed fiducial carries it. The conversion, with the bracketed quantities DESI chain values and σ_8 at the DESI fiducial cosmology, is

$$b_1 = \frac{[(1+b_1^{\rm L})\sigma_8]}{\sigma_8(z_{\rm eff})}, \quad b_2 = \frac{[b_2^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} + \frac{8}{21}\,(b_1-1)\,, \quad b_{s^2} = \frac{2\,[b_s^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} - \frac{4}{7}\,(b_1-1)\,. \tag{10}$$

Applied to the draft's numbers it gives $b_s^{\rm L} \simeq 2.5$–3.3 and $b_{s^2} \simeq +4$ to $+6$: chain values dominated by the width of the DESI prior and by degeneracies with the counterterms, not a measurement of the tidal field. The model therefore imports no DESI nuisance value: its fiducials are the calibrated relations of Section 3.3, and the DESI posteriors motivate only its prior widths.

6 The size of every term

Figure 3 draws each one-loop term of Eq. (2) in percent of the linear-bias model $b_1^2 P_{\rm mm}$, for four DESI-like tracers at the calibrated biases of Figure 1 (Appendix C gives the setup). Figure 4 shows, on the same scale, the Poisson power of the project's thin bins, which the covariance carries as noise and which sets the scale of the stochastic freedom.

At the exploration variant's cut. Each bin is cut at $0.075\,h\,{\rm Mpc}^{-1}/D(\bar{z}_i)$: 0.088, 0.097, 0.112 and $0.139\,h\,{\rm Mpc}^{-1}$ for the four tracers of the figure. At its own cut the loop sum of each tracer is +1.8% (BGS-like), +2.2% and +6.7% (LRG-like at $z = 0.5$ and 0.8) and −0.4% (ELG-like, where terms of ±4% cancel); the largest single term is the $b_1 b_2$ loop in every case. The figure keeps the low-k constants. In CosmoLike's subtracted convention the BGS-like and ELG-like sums become −5% and −6% and the LRG-like ones move by under a point; the difference is white noise, carried by the subtraction and by SN_0, not a change of shape. The $b_1 b_{3\rm nl}$ term stays below 0.3% at $k \le 0.1\,h\,{\rm Mpc}^{-1}$. Thirty percent of the Poisson power, the non-Poisson stochasticity Nicola et al. [8] detect in red samples, is 4–5% of P_{gg} for the LRG-like tracers at their cuts and 31% for the ELG-like tracer, whose Poisson power there exceeds its clustering power. Against the forecast's sub-percent precision on each group's bias amplitude (the correlated-prior note's forecasts), none of these is negligible. Below the cut their scale dependence is mild, so the linear amplitudes absorb most of each term and what remains is the residual shape: the case for calibration-centered freedom with modest priors, the exploration variant.

At the shipped mask. The fiducial widths' reference cut, the shipped mask, stops every bin at $k = 0.176\,h\,{\rm Mpc}^{-1}$. There the $b_1 b_2$ term alone is 4–14% of P_{gg}, with strong scale dependence, the loop sum reaches 15% (LRG-like at $z = 0.8$), and the counterterm at DESI's width is 6–15%. No amplitude absorbs these terms, and their shapes differ from one another: the case for the fiducial widths' wider offsets and the radial counterterm, the freedom DESI DR1 carries with seven nuisances per tracer at $k_{\rm max} = 0.2\,h\,{\rm Mpc}^{-1}$.

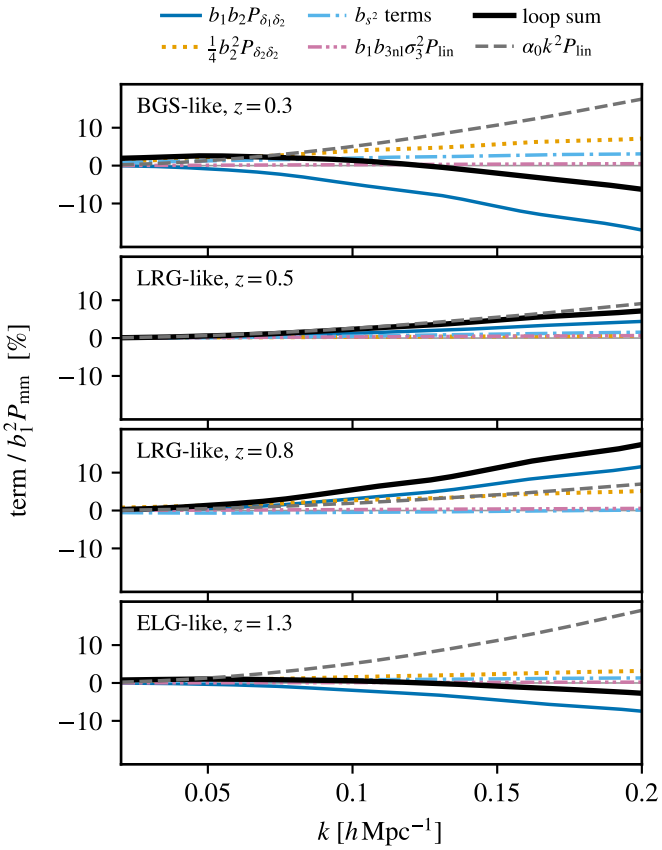


Figure 3: Each one-loop term of Eq. (2) in percent of the linear-bias model $b_1^2 P_{\mathrm{mm}}$ (halofit), against k at the tracer redshift, for four DESI-like tracers at the calibrated biases of Figure 1. The three tidal terms ($b_1 b_{s^2} P_{\delta_1 s_2}$, $\frac{1}{2} b_2 b_{s^2} P_{\delta_2 s_2}$ and $\frac{1}{4} b_{s^2}^2 P_{s_2 s_2}$) are drawn as their sum. Loop templates are FAST-PT at $z = 0$ scaled by $D^4(z)$, with the low-k constants kept (CCL's default, `sub_lowk` off). The dashed curve is the counterterm $\alpha_0 k^2 P_{\mathrm{lin}}(k, z)$ at DESI's width, $\alpha_0 = 12.5\, h^{-2}\mathrm{Mpc}^2$.

7 The model as implemented

7.1 One likelihood option, every term per bin

The DESI likelihood (`likelihood/desi_cl.py`) runs the one-loop model when its option `higher_order_bias` is 1 (0 is linear bias). Every term is per bin. The galaxy power of bin i in redshift space is

$$P_{g,i}(k,\mu) = P_{gg,i}(k) + \left(2\, b_{1,i}\, f\mu^2 + f^2\mu^4\right) P_{\mathrm{lin}}(k) + \left(b_{1,i} + f\mu^2\right) f\, \alpha_{2,i}\, \mu^2 k^2\, P_{\mathrm{lin}}(k) + \frac{1}{\bar{n}_i}\left(\mathrm{SN}_{0,i} + \mathrm{SN}_{2,i}\, f_{\mathrm{sat}} \sigma_{\mathrm{sat}}^2\, k^2 \mu^2\right), \tag{11}$$

whose real-space part is Eq. (2) in the subtracted convention, with $b_{3\mathrm{nl}} = 0$ and the counterterm written on P_{mm}:

$$P_{gg,i}(k) = b_{1,i}^2\, P_{\mathrm{mm}}(k) + P_{\mathrm{loop}}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) + \alpha_{0,i}\, k^2\, P_{\mathrm{mm}}(k), \tag{12}$$

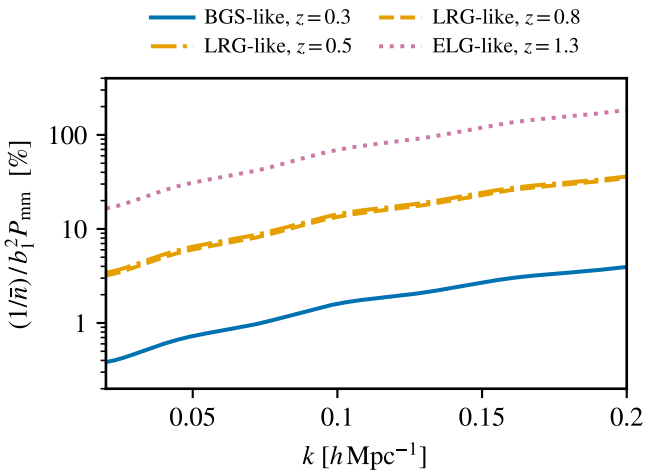


Figure 4: The Poisson power $1/\bar{n}$ in percent of the linear-bias model $b_1^2 P_{\rm mm}$ (halofit) of the four tracers of Figure 3, with $\bar{n}$ the density of the project's thin bin nearest each tracer redshift. The covariance carries this power as noise; it is drawn to set the scale of the stochastic freedom.

with $P_{\rm loop}$ the five quadratic templates of Eq. (2). Redshift-space distortions are linear (Kaiser) apart from the α_2 term, and each bin's redshift distribution is convolved with the sampled Fingers-of-God dispersion σ_v, as in the linear-bias model. The terms:

- **Quadratic and tidal biases** follow each bin's own linear bias through the calibrations of Section 3.3, plus offsets:

$$b_{2,i} = b_2^{\rm fit}(b_{1,i}) + \Delta_{2,i}, \qquad b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1) + \Delta_{s,i}. \tag{13}$$

  Section 7.2 explains why the relations take the full $b_{1,i}$ and how the offsets are grouped.

- $b_{\rm 3nl} = 0$, DESI's baseline. The co-evolution value it discards, $|b_{\rm 3nl}| \leq 0.17$, is worth at most 0.3% of P_{gg} at $k \leq 0.1\,h\,{\rm Mpc}^{-1}$ (Figure 3) and is close in shape to the counterterm; CosmoLike's $b_{\rm 3nl}$ template is also unusable (Appendix D).

- **The counterterm** α_0, in $h^{-2}{\rm Mpc}^2$, rides CosmoLike's derivative-bias slot $2\,b_1 b_K k^2 P_{\rm mm}$, with $b_K = \alpha_0/[2b_1(c/H_0)^2]$ and $c/H_0 = 2997.92\,h^{-1}{\rm Mpc}$, because CosmoLike measures k in units of H_0/c. The term multiplies $P_{\rm mm}$, where DESI's α_0 multiplies $(b_1 + f\mu^2)b_1 P_{\rm lin}$: the two coefficients differ by a factor b_1^2 and by the ratio $P_{\rm mm}/P_{\rm lin}$, at most 1.16 at $k \leq 0.2\,h\,{\rm Mpc}^{-1}$ for the tracers of Figure 3.

- **The radial counterterm** α_2, in $h^{-2}{\rm Mpc}^2$, follows the DESI DR1 field convention [5]: the galaxy density gains $\frac{1}{2} f\mu^2 \alpha_2 k^2 \delta_{\rm L}$, with $\delta_{\rm L}$ the linear density contrast, and twice its cross with the Kaiser field $(b_1 + f\mu^2)\delta_{\rm L}$ is the α_2 term of Eq. (11). The likelihood passes $\alpha_{2,i}$; the C code (`galpha2` in `bias.c`) applies $b_{1,i}$ and f.

- **The white noise** SN_0, in units of the bin's Poisson power $1/\bar{n}_i$: the deviation from Poisson, since the covariance carries the Poisson part. $\bar{n}_i$ is fixed by the bin's surface density, so that $1/\bar{n}_i$ projected with the raw redshift distribution is the angular shot noise.

- **The line-of-sight stochastic term** $\mathrm{SN}_2 k^2\mu^2$, in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$, with the DESI DR1 values [5] $(f_{\rm sat}, \sigma_{\rm sat}) = (0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG and $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG; $\sigma_{\rm sat}$ is not the Fingers-of-God σ_v. BGS, absent from DESI's table, takes the LRG values, a choice without consequence (Section 8.3).

The compiled code has no stochastic slot, so the likelihood adds SN_0 and SN_2 as exact templates (Section 8.3). A zero amplitude switches a term off at no cost: $\alpha_2 = 0$ in every bin skips the term in the C code, and a stochastic family whose amplitudes all vanish is not computed. Of Eq. (8) the model drops α_4 and SN_4, which DESI DR1 drops with the hexadecapole, and the redshift-space one-loop terms, which CosmoLike does not have (Appendix D); the derivative bias b_{∇^2} is exactly degenerate with α_0 at one loop.

Why α_2 takes DESI's field convention. In that convention $\alpha_2 k^2$ is a fractional correction: the α_2 term is $\alpha_2 k^2$ times the share of the Kaiser power carried by the velocity term, $(b_1 + f\mu^2) f\mu^2 P_{\rm lin}$, whatever the bias and the growth rate. A prior width of $0.5/k_{\rm max}^2$ is then a 50% correction at $k = k_{\rm max}$ in every bin, the meaning of DESI's $12.5\,h^{-2}\mathrm{Mpc}^2$ at $0.2\,h\,\mathrm{Mpc}^{-1}$. Written as a bare power, $\alpha_2\mu^2 k^2 P_{\rm lin}$, the same width would be a correction whose size changes with $b_{1,i}$ and f from bin to bin.

7.2 Bias evolution within a group

The linear sector is the per-bin bias construction of the correlated-prior note (its "memory model"): one amplitude per group, with each bin free to deviate under a correlated prior. Bin i of group g has

$$b_{1,i} = \frac{B_g\,(1+\delta_i)}{D(\bar{z}_i)}, \qquad \delta_i = s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{14}$$

with B_g the group amplitude, $\bar{z}_g$ the mean redshift of the group's bins, δ_i the fractional deviation from the growth-tied trend B_g/D (one number per bin, not the matter overdensity δ), s_g a free tilt, and ε_i a residual under a Gaussian prior correlated within the group,

$$\left\langle \varepsilon_i\,\varepsilon_j \right\rangle = \sigma_g^2\,\exp\!\left(-\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{15}$$

and zero across groups; σ_g is the per-bin width and L the correlation length in redshift. The four linear-bias priors are growth-tied ($\delta_i = 0$), baseline ($\sigma_g = 3\%$ for BGS and 1% elsewhere, $L = 0.2$), pessimistic (3% everywhere, $L = 0.05$) and free (one flat bias per bin); the correlated-prior note derives them.

The calibrations of Eq. (13) take the full $b_{1,i}$, deviations included, not the smooth trend $B_g/D(\bar{z}_i)$. A deviation δ_i is a real change of which halos the selection reaches in that bin (a flux limit keeps only the brighter galaxies, which sit in more massive halos), and the calibrations map halo populations at a given linear bias, so the second-order biases of the bin's population sit on the calibration curves at $b_{1,i}$. The induced shift is small but determined: near $b_1 \simeq 2$ the Lazeyras relation has slope $db_2/db_1 \simeq 1.7$, so a 3% residual moves b_2 by about 0.10, a tenth of the exploration variant's offset width, and the co-evolution slope $-4/7$ moves b_{s^2} by about 0.03.

The deterministic part carries the within-group evolution, which is not small: across LRG3 the co-evolution b_{s^2} moves from -0.88 to -1.09 (Figure 1), a drift comparable to the offset width. Holding b_2 and b_{s^2} constant across a group while $b_{1,i}$ evolves would be inconsistent. Freeing them per bin, 152 parameters on the 62 Mpc layout, is a bound rather than a model: a

per group	exploration variant	fiducial widths
offsets Δ_2 / Δ_s	1.0 / 0.5	2.0 / 1.0
α_0 $[h^{-2}\mathrm{Mpc}^2]$	$0.5/k_{\mathrm{max}}^2$	$0.5/k_{\mathrm{max}}^2$
α_2 $[h^{-2}\mathrm{Mpc}^2]$	fixed to 0	$0.5/k_{\mathrm{max}}^2$
SN_0 $[1/\bar{n}_i]$	2	2
SN_2 $[f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2/\bar{n}_i]$	5	5
tilted (+ tilts only)	Δ_2, Δ_s, α_0	Δ_2, Δ_s, α_0, α_2
parameters, without / with tilts	5 / 8	6 / 10

Table 1: The widths of the Gaussian priors of the higher-order parameters, all centered on zero, and their number per group (the tilts, with flat priors, only in groups of more than one bin). k_{max} is the fit's cut, quoted at $z = 0$; each bin is cut at $k_{\mathrm{max}}/D(\bar{z}_i)$. The exploration variant: $k_{\mathrm{max}} = 0.075\,h\,\mathrm{Mpc}^{-1}$, an α_0 width of $88.9\,h^{-2}\mathrm{Mpc}^2$. The fiducial widths: each cut of a scan, or the shipped mask, $0.176\,h\,\mathrm{Mpc}^{-1}$ in every bin ($16.1\,h^{-2}\mathrm{Mpc}^2$).

bin's loop terms are nearly degenerate with its linear bias and counterterm, so the prior would set the per-bin posteriors.

What a free parameter adds on top of the inherited trend is a drift of the calibration itself: b_2 at fixed b_1 changes with the satellite fraction and the color mix, slowly within one selection. The freedom therefore sits per group. Both width configurations give each group one offset, $\Delta_{2,i} = \Delta_{2,g}$ (and likewise Δ_s); their *tilted* variants add a free straight line in redshift within the group, the kind of line s_g gives the linear bias, $\Delta_{2,i} = \Delta_{2,g} + t_{2,g}(\bar{z}_i - \bar{z}_g)$, with a flat prior on the tilt $t_{2,g}$. The counterterms follow the same pattern, because their scale follows the size of the host halos, a population property that drifts with the selection. The stochastic amplitudes never tilt: the Poisson part $1/\bar{n}_i$ is known per bin and sits in the covariance, so only the non-Poisson fraction could drift, and only what has a physical driver tilts.

7.3 The exploration variant and the fiducial widths

The fiducial widths are shipped as one parameter file of the one option per layout, written by `scripts/make_desi_params.py --model-b`; "model b" in the flag and the file names is the historical key of the fiducial widths. Each file, `likelihood/params_desi[_<case>]_model_b_tilts.yaml`, states the offsets and tilts of Table 1, with α_2 fixed to 0 in every bin, the paper's choice. The paper yamls `EXAMPLE_EVALUATE1–4.yaml` run them on the four layouts. The exploration variant, built for $k_{\mathrm{max}} = 0.075\,h\,\mathrm{Mpc}^{-1}$, is not shipped; its column in Table 1 is the record of the narrower configuration that motivated the fiducial widths. The likelihood reads every parameter per bin; a group is a yaml statement that ties its bins to one offset, or to an offset and a tilt. Either width configuration, with or without tilts, runs under any of the linear-bias priors.

The widths have stated sources.

- **Offsets.** The exploration variant's 1.0 and 0.5 cover the known imperfection of the calibrations: the scatter of the Lazeyras fit and the 30% spread between co-evolution and the excursion-set b_{s^2} at LRG biases. The fiducial widths double them, demoting the calibrations to prior centers for cuts where the loops reach 15% of P_{gg}.

- **Counterterms.** The DESI rule of a 50% correction at the cut, evaluated at each cut: $0.5/k_{\mathrm{max}}^2$, which is DESI's 12.5 at $0.2\,h\,\mathrm{Mpc}^{-1}$. Because α_0 multiplies P_{mm}, at bin i's own

cut the width is a correction of $0.5/B_g^2$ of the linear galaxy power $b_{1,i}^2 P_{\rm mm}$, the same in every bin of a group: 28% for BGS, 17% for LRG, 71% for ELG at the exploration variant's cut.

- **Stochastic terms.** DESI DR1's widths.

The exploration variant is built for $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$, the cut matched to the thin-bin radial resolution. It fixes $\alpha_2 = 0$: at DESI's width the α_2 term changes the kept auto spectra by about 1% (Section 8.2), and a fit without α_2 holds against a truth with $\alpha_2 = \pm 12.5\,h^{-2}{\rm Mpc}^2$ (`notes/fog_parameter_degeneracy_study`). It carries $\rm SN_2$ because without it the thin layouts fail at its cut (Section 8.3). The fiducial widths free α_2 for the scan in $k_{\rm max}$. With the amplitude B_g and the linear tilt s_g, the fiducial-width fit has eight parameters per group: DESI DR1's seven per tracer [6, 7] plus the linear tilt the thin bins resolve. Every fiducial higher-order parameter is zero, so the fiducial model sits on the calibrations; no DESI chain value enters (Section 5).

7.4 The paper model

The paper forecast runs one configuration of this option – the fiducial model: the fiducial widths with their tilts, α_2 fixed to zero in every bin and every σ_v prior width multiplied by six, on the pessimistic linear-bias prior. Its example yamls are `EXAMPLE_EVALUATE1`–`4`, built on the parameter files `params_desi[_<case>]_bias_pessimistic.yaml` (the pessimistic linear bias) and `params_desi[_<case>]_model_b_tilts.yaml` (the fiducial model's higher-order sector; "model b tilts" is the file's historical key). The sampled Fingers-of-God dispersions reproduce the α_2 derivative templates almost exactly, so dropping α_2 and widening the σ_v priors sixfold lets the dispersions absorb an α_2 of DESI's size: fitted to a truth with $\alpha_2 = \pm 12.5\,h^{-2}{\rm Mpc}^2$, the paper model moves no scored parameter (the cosmological parameters the forecast reports) by more than 0.33σ (`notes/fog_parameter_degeneracy_study` derives the factor of six and the shifts). Every other width is the fiducial one of Table 1; Section 2 lists the σ_v fiducials and widths.

7.5 The covariance

The Gaussian covariance is built on the one-loop fiducial: $b_{1,i} = B_g/D(\bar{z}_i)$, b_2 and b_{s^2} on the calibrations, every other term zero. Both width configurations, with or without tilts, share that fiducial, so one covariance serves all of them. Its one-loop terms come from CosmoLike's own tables of C-FASTPT (its C port of FAST-PT), with the data vector's bilinear pair form (Section 8.1), and enter through Limber; the low-ℓ non-Limber recompute stays linear (Appendix D). `nonlimber_accuracyboost` 2 converges the adjacent thin-bin cross spectra to 2% and the diagonal to 0.4%. The covariance takes massless neutrinos, so its signal sits 3% above the data vector ($m_\nu = 0.06\,{\rm eV}$) at every redshift. Against a linear-bias covariance, the figures of merit (FoM: the inverse area of a parameter pair's marginalized error ellipse; `notes/fisher_machinery`) move by -0.4% to $+1.2\%$.

8 Projection to thin-bin angular spectra

8.1 Separability and the D^4 rule

The exact (non-Limber) angular power spectrum between bins i and j is

$$C_\ell^{ij} = \frac{2}{\pi} \int dk\, k^2 \int d\chi_1\, W_i(\chi_1)\, j_\ell(k\chi_1) \int d\chi_2\, W_j(\chi_2)\, j_\ell(k\chi_2)\, \mathcal{P}(k; z_1, z_2)\,, \tag{16}$$

where $W_i(\chi)$ is the bin's radial kernel (the normalized, Fingers-of-God-convolved redshift distribution expressed in comoving distance χ), j_ℓ is the spherical Bessel function, and $\mathcal{P}$ is the unequal-time galaxy power spectrum. Every term of the one-loop expansion is separable in k and time:

- the linear term carries $D(z_1)\, D(z_2)$;
- the five quadratic templates are 22-type mode-coupling integrals over two linear spectra, so each carries $D^2(z_1)\, D^2(z_2)$;
- the 13-type pieces (the $\sigma_3^2 P_{\rm lin}$ term of $b_{\rm 3nl}$ and the P_{13} inside the one-loop matter spectrum) carry the symmetrized $\frac{1}{2}\left[D^3(z_1)D(z_2) + D(z_1)D^3(z_2)\right]$.

At equal times every loop term scales as $D^4(z)$. CCL, velocileptors and CosmoLike all compute the templates once from the $z = 0$ linear spectrum and rescale them by D^4 [7]; CosmoLike's `get_FPT_bias` (`pt_cfastpt.c`) builds them with C-FASTPT and rebuilds them when the cosmology changes. For thin bins the equal-time evaluation at the bin mean is accurate: across a $\Delta z = 0.02$ bin the growth factor changes by about 1%, the symmetric products differ from $D^4(z_{\rm mid})$ only at second order in that 1%, and the residual is a few parts in 10^4 of the loop terms.

The auto spectrum of Eq. (2) is a quadratic form in the bias vector $u_i = (b_{1,i}, b_{2,i}, b_{s^2,i})$, $P_{gg,i} = u_i^{\mathsf{T}} Q\, u_i$, with Q the symmetric matrix of templates ($Q_{11} = P_{\rm mm}$, $Q_{12} = \frac{1}{2}P_{\delta_1\delta_2}$, $Q_{22} = \frac{1}{4}P_{\delta_2\delta_2}$, and so on). A cross pair of bins i and j carries the bilinear form $u_i^{\mathsf{T}} Q\, u_j$: each product of two biases is shared between the two bins. The low-rank engine – the project's non-Limber method for galaxy clustering, which writes the power spectrum as a short sum of terms separable in k and redshift and projects each term exactly (the correlated-prior note builds and tests it) – projects each bin's full one-loop $P_{gg,i}$, counterterm included, through a singular-value decomposition (SVD) of its square root, so it carries the one-loop sum with no structural change; for the cross pairs it uses the bilinear form.

CosmoLike subtracts the low-k constants of the quadratic templates (Section 3.1), so the loop sum vanishes as $k \to 0$ and large scales keep the linear-bias spectrum. As $k \to \infty$ the templates decay and the subtraction leaves a constant,

$$P_\infty = -\frac{\sigma_4}{2}\, D^4(z) \left(b_2 + \frac{2}{3}\, b_{s^2}\right)^2 \leq 0, \tag{17}$$

with $\sigma_4 = 4391\,(h^{-1}{\rm Mpc})^3$ at $z = 0$ at the likelihood's fiducial. Where this negative constant outweighs the decaying positive terms, P_{gg} turns negative at high k and its square root does not exist. The low-rank engine handles the constant exactly: it factorizes $P_{gg} - P_\infty$ and projects P_∞ separately (Section 8.4).

8.2 Linear redshift-space distortions and the counterterms

Linear (Kaiser) redshift-space distortions enter Eq. (16) through the transformation of the density kernel: each factor of $f\mu^2$ in the flat-sky spectrum becomes a second derivative of the Bessel function in the exact projection, because $\mu^2 = (k_\parallel/k)^2$ and each radial derivative of the projected plane wave brings down one power of $k_\parallel$ – two of them acting on $j_\ell(k\chi)$ give j_ℓ'':

$$\Delta_\ell^i(k) = \int d\chi\, W_i(\chi)\left[b_1(z)\, j_\ell(k\chi) - f(z)\, j_\ell''(k\chi)\right] D(z),$$
$$R_\ell^i(k) = -\int d\chi\, W_i(\chi)\, f(z)\, j_\ell''(k\chi)\, D(z), \tag{18}$$

with Δ_ℓ^i the whole linear transfer of bin i and R_ℓ^i its velocity part. For wide photometric bins the j_ℓ'' term integrates down by orders of magnitude; for shells thinner than the scales probed it survives at full strength, because the shell resolves line-of-sight modes with μ near one. This is why thin-bin C_ℓ^{gg} recovers the Kaiser information of a three-dimensional analysis [10], and it cuts both ways: every μ-dependent term that a wide-bin analysis could drop is alive here.

α_0. The α_0 term is μ-independent and projects like the density: one extra k-shape, $k^2 P_{\rm mm}$, in each bin's $P_{gg,i}$. The engine matters. FKEM, the method of Fang et al. [11] built on FFTLog (the fast Bessel transform on a logarithmic grid), projects only the linear term exactly and moves every nonlinear term, the counterterm included, through its Limber difference; the low-rank engine projects the counterterm without the Limber approximation. On the $\Delta z = 0.05$ layout the low-rank α_0 response is 7.3 times the Limber one at $\ell = 20$ and 1.26 times at $\ell = 100$, from the k^2-weighted radial modes of thin, sharp-edged bins, so the model runs on the low-rank engine (the correlated-prior note, `notes/correlated_bias_prior_v2`, on why the counterterm requires the low-rank engine).

α_2. The counterterm field of bin i, $\frac{1}{2} f\mu^2 \alpha_{2,i} k^2 \delta_{\rm L}$, has the transfer $\frac{1}{2}\alpha_{2,i} k^2 R_\ell^i$, and its cross with the linear field of bin j adds

$$\Delta C_\ell^{ij} = \frac{2}{\pi}\int dk\, k^2\, P_{\rm lin}(k)\, \frac{k^2}{2}\left[\alpha_{2,i}\, R_\ell^i\, \Delta_\ell^j + \alpha_{2,j}\, \Delta_\ell^i\, R_\ell^j\right], \tag{19}$$

which for an auto spectrum is $\alpha_{2,i} k^2 \Delta_\ell^i R_\ell^i$ under the integral. The projection reuses the two transfers the Kaiser term already builds, adds no Bessel integral, and is exact; every gg engine carries it (FFTLog, low-rank, FKEM pairs, and Limber with $P_{\rm lin}$). It is linear in α_2 to 10^{-12}, and a brute-force j_ℓ, j_ℓ'' double integral agrees with it to 6×10^{-4} for a thin auto spectrum and an adjacent pair at $\ell = 2$–500. At DESI's width, $12.5\, h^{-2}{\rm Mpc}^2$, it raises the auto spectra of the shipped mask by a median 0.7% ($\Delta z = 0.05$) and 1.1% ($\Delta d_{\rm com} = 62\,{\rm Mpc}$), at most 2.9%.

The Fingers-of-God convolution of each bin's redshift distribution is a nonperturbative stand-in for part of the same physics: it damps radial power with a shape set by the sampled σ_v, and is partially degenerate with α_2. Partial degeneracy is not identity: the convolution is a positive damping of fixed functional form, while α_2 takes either sign and scales as $k^2 P_{\rm lin}$.

8.3 Stochastic terms in projection

A constant three-dimensional power N projects to

$$C_\ell^{ij}\big|_{\rm stoch} = N\int d\chi\, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2}, \tag{20}$$

independent of ℓ. Equation (20) is exact, not a Limber approximation: for a constant $\mathcal{P}$ the k integral of Eq. (16) collapses by Bessel orthogonality, $\int dk\, k^2 j_\ell(k\chi_1)\, j_\ell(k\chi_2) = \frac{\pi}{2\chi_1^2}\,\delta_D(\chi_1-\chi_2)$ with δ_D the Dirac delta, so the Limber formula is exact for white noise. For a top-hat bin of width $\Delta\chi$ the result is $N/(\chi^2\Delta\chi)$, and at the Poisson value $N = 1/\bar{n}$ it reproduces the angular shot noise $1/\bar{n}_{2D}$, with $\bar{n}_{2D}$ the surface density of the bin. The model's SN_0 templates are Eq. (20) with the Fingers-of-God-convolved kernels, which overlap for adjacent thin bins, so a stochastic term also reaches the measured cross spectra. A cross pair within one tracer sample takes the mean of the two bins' Poisson powers; a pair of two samples (an LRG and an ELG bin at the same redshift) takes none, because different galaxies share no stochastic term.

SN_2 **projects exactly too.** A power proportional to $k^2\mu^2 = k_\parallel^2$ is that of the radial derivative $\partial_\chi\epsilon$ of a white noise ϵ. Projected with the kernel of bin i, $\int d\chi\, W_i\, \partial_\chi\epsilon = -\int d\chi\, W_i'\,\epsilon$ (integration by parts, with $W_i' = dW_i/d\chi$ and the kernel vanishing at its ends), so the SN_2 term is Eq. (20) with each kernel replaced by its radial derivative:

$$C_\ell^{ij}\big|_{\mathrm{SN}_2} = U_{ij}\int d\chi\,\frac{W_i'(\chi)\,W_j'(\chi)}{\chi^2}, \qquad U_i = \frac{\mathrm{SN}_{2,i}\, f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2}{\bar{n}_i}, \tag{21}$$

with U_{ij} the mean of U_i and U_j within one sample and zero across samples. A thin bin's kernel is a top hat smoothed only by the Fingers-of-God convolution, so W_i' peaks at the two edges, positive on the rising edge and negative on the falling one. The sharper the edges, the larger the template: per unit, the SN_2 auto template is a median 0.022, 0.017, 0.0065 and 0.0017 times the SN_0 one on the 62 Mpc, $\Delta z = 0.02$, 0.05 and 0.2 layouts.

Neighbors are anticorrelated. Adjacent thin bins share an edge: the falling edge of one is the rising edge of the next, so $W_i'W_j' < 0$ there and the projected SN_2 of two neighbors is negative. On the 62 Mpc layout the normalized adjacent term, $K_{ij}/\sqrt{K_{ii}K_{jj}}$ with K_{ij} the integral of Eq. (21), has a median of -0.50 over the 73 adjacent pairs, against $+0.05$ for SN_0. A white noise adds only positive cross terms, products of positive kernels, so no SN_0, per group or per bin, reproduces the pattern.

Why the model carries SN_2**.** Without SN_2, a truth with $\mathrm{SN}_2 = \pm 5$ units, the one-sigma width of its prior, moved an exploration-variant fit's Ω_m by 1.3–1.7σ at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ on the two thin layouts, mostly through the adjacent cross spectra. The fit then held only to $0.05\,h\,\mathrm{Mpc}^{-1}$ on the 62 Mpc layout and failed at its first fitted cut, $0.075\,h\,\mathrm{Mpc}^{-1}$, on $\Delta z = 0.02$. The LRG term carried the effect. ELG alone moved Ω_m by at most 0.23σ (its unit, $0.84\,(h^{-1}\mathrm{Mpc})^2$, is a seventh of LRG's 6.08), and BGS by 0.00, with the borrowed LRG unit or with one built from the BGS bins' own Fingers-of-God dispersion, so the BGS unit does not matter. With one SN_2 per group, the exploration variant holds to its own cut on the thin layouts. About half of each bin's SN_2 auto kernel comes from $k_\parallel$ above the bin's own cut at $k_{\max} = 0.075\,h\,\mathrm{Mpc}^{-1}$, a content the scale cut does not weigh (Appendix D).

8.4 The white-noise limit of the renormalized loops

Origin of the constant. The template $P_{\delta_2\delta_2}$ is the power spectrum of the operator δ^2. Its four linear fields pair across the two operators in two ways (two Wick contractions), and each pairing gives $\int_{\mathbf{q}} P_{\mathrm{lin}}(q)\, P_{\mathrm{lin}}(|\mathbf{k}-\mathbf{q}|)$. As $k \to 0$ this integral tends to $\sigma_4 = \int_{\mathbf{q}} P_{\mathrm{lin}}^2(q)$, so $P_{\delta_2\delta_2} \to 2\sigma_4$. The

constant comes from pairs of nearly opposite wavevectors, $\mathbf{q}$ and $\mathbf{k}-\mathbf{q}\simeq-\mathbf{q}$, whose sum is the long mode; on scales much larger than $1/q$ it is indistinguishable from white noise. The tidal templates carry the kernel $S_2=\mu_{12}^2-\frac{1}{3}$ of Appendix B, which equals $\frac{2}{3}$ for opposite wavevectors ($\mu_{12}=-1$): one power of it gives $P_{\delta_2 s_2}\to\frac{4}{3}\sigma_4$, two give $P_{s_2 s_2}\to\frac{8}{9}\sigma_4$.

Subtracting these constants renormalizes the operators: the constant moves into the stochastic sector and the loops vanish on large scales. As $k\to\infty$ the templates decay, so the subtracted loop sum of Eq. (2) tends to minus the subtracted constants, each with its bias coefficient:

$$\tfrac{1}{4}b_2^2\,(2\sigma_4)+\tfrac{1}{2}b_2 b_{s^2}\,(\tfrac{4}{3}\sigma_4)+\tfrac{1}{4}b_{s^2}^2\,(\tfrac{8}{9}\sigma_4)=\frac{\sigma_4}{2}\left(b_2+\tfrac{2}{3}b_{s^2}\right)^2. \tag{22}$$

With the D^4 scaling, this square with a minus sign is P_∞ of Eq. (17). The perfect square has a reading: $b_2+\frac{2}{3}b_{s^2}$ is the quadratic coefficient in the Galileon basis of Appendix A, and the Galileon kernel $S_2-\frac{2}{3}=\mu_{12}^2-1$ vanishes for opposite wavevectors, so $\mathcal{G}_2$ adds no constant of its own.

Two conventions, one model. CosmoLike subtracts the three constants, as CCL does with `sub_lowk` on. Velocileptors, in both its Eulerian and Lagrangian flavors, keeps the loop proportional to the square of the quadratic bias unsubtracted (in the Lagrangian flavor $\frac{1}{2}(b_2^{\rm L})^2\xi_{\rm lin}^2$ in configuration space, with $\xi_{\rm lin}$ the linear correlation function); it removes only the operator means of Eq. (3), which touch only the $k=0$ mode, and its SN_0 absorbs the low-k constant [4, 5]. The subtracted spectrum with SN_0 centered at zero equals the unsubtracted spectrum with SN_0 centered at $-(\sigma_4/2)D^4(b_2+\frac{2}{3}b_{s^2})^2$, a center that moves with the biases. A prior on SN_0 with a fixed center therefore states different physics in the two conventions.

Where P_{gg} turns negative. Let $k_{\rm neg}$ be the wavenumber at which the subtracted one-loop P_{gg} first turns negative, and $\Delta_2=b_2-b_2^{\rm fit}(b_1)$ the offset of Eq. (13). Figure 5 draws $k_{\rm neg}$ against Δ_2 for three representative bins. In the LRG bins the constant stays small against the galaxy power, because the calibrated b_2 partly cancels $\frac{2}{3}b_{s^2}$ and the larger b_1^2 raises the positive terms: $k_{\rm neg}>1.5\,h\,\mathrm{Mpc}^{-1}$. At the calibrated biases ($\Delta_2=0$), $|P_\infty|$ is 5–42 times the Poisson power in the forecast's BGS bins, where $1/\bar{n}\simeq 33$–$170\,(h^{-1}\mathrm{Mpc})^3$, and 3–15% of it in the ELG bins. Only negative offsets matter: $b_2+\frac{2}{3}b_{s^2}$ vanishes near $\Delta_2\simeq+0.9$, so a positive offset shrinks the constant and pushes $k_{\rm neg}$ far out.

Why thin-bin angular spectra see it and DESI's fits do not. An angular spectrum integrates $P(k)$ over all k through each bin's radial window (Eq. (16)). A thin bin passes radial modes far above the transverse wavenumber of a band, so no cut in ℓ excludes the region $k>k_{\rm neg}$; the UV-fraction cut only bounds its share. A white noise projects exactly (Eq. (20)), so in the subtracted convention P_∞ enters every multipole at full strength. DESI's full-shape fits are spared both effects. They use only $0.02<k<0.20\,h\,\mathrm{Mpc}^{-1}$ in three dimensions [6], and the DESI DR1 samples are sparse: DR1 BGS holds 300,017 galaxies over $0.1<z<0.4$, which with the survey area gives $1/\bar{n}\approx 3\times10^3\,(h^{-1}\mathrm{Mpc})^3$ (an estimate made for this note). The constant is then at most of the order of the Poisson power and fits DESI's SN_0 prior in either convention. The forecast's BGS is 20–100 times denser.

What CosmoLike does. The low-rank engine splices its non-Limber density term into the Limber spectrum: it takes the full Limber C_ℓ, subtracts the Limber projection of the density-

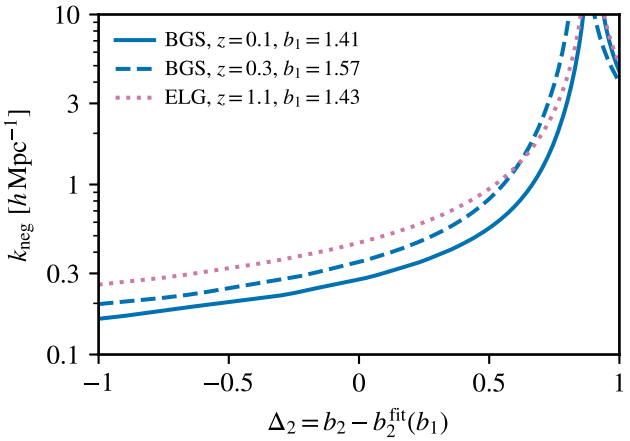


Figure 5: The wavenumber $k_{\rm neg}$ at which the one-loop P_{gg} in CosmoLike's subtracted convention first turns negative, against the offset Δ_2 of b_2 from the Lazeyras calibration, for two BGS bins and one ELG bin. b_{s^2} is on co-evolution, $b_1 = B_g/D(z)$, and $b_{\rm 3nl}$ and the counterterm are zero. The leading term is $b_1^2 D^2 P_{\rm lin}$; the halofit $P_{\rm mm}$ of the model is larger at these wavenumbers, so the model turns negative at somewhat higher k. Where $b_2 + \frac{2}{3}b_{s^2}$ vanishes, P_∞ vanishes with it and $k_{\rm neg}$ leaves the panel. Setup in Appendix C.

only term, and adds the non-Limber projection of the same term. The white-noise constant passes through this splice exactly:

- the engine factorizes $P_{gg} - P_\infty$, the spectrum with the unsubtracted templates, which is positive, so its square root exists;
- the Limber density-only term subtracted in the splice uses the same unsubtracted templates;
- the full Limber term keeps the subtracted templates, so P_∞ reaches C_ℓ through Limber, exact for a constant power;
- a cross pair carries the bilinear constant $-(\sigma_4/2)D^4(b_{2,i} + \frac{2}{3}b_{s^2,i})(b_{2,j} + \frac{2}{3}b_{s^2,j})$.

A floor on the factorized spectrum stays in the code as a guard; the tested vectors do not reach it. Compared with flooring P_{gg} instead, the exact treatment leaves linear-bias vectors and FKEM one-loop vectors bitwise unchanged, as it must, and moves the low-rank one-loop vector at the calibrated biases by at most 5×10^{-3} of the angular shot noise. At the highest band the exploration variant keeps, the share of a bin's white-noise projection from $k > k_{\rm neg}$ is at most 0.34% (BGS, $z = 0.11$) and about zero for ELG (a flat-sky estimate at the calibrated biases). One exploration-variant prior width below the calibrations ($\Delta_2 = -1$, $\Delta_s = -0.5$), the exact treatment shows negative C_ℓ in the lowest BGS bins beyond $k \approx 0.15\,h\,{\rm Mpc}^{-1}$ (Limber correspondence $k = (\ell + \frac{1}{2})/\chi$) on the shipped mask, -29 times the shot noise at $\ell = 50$ (-25 with FKEM, Limber in its one-loop content), which a floor would hide. At the exploration variant's cut the same biases keep every kept C_ℓ positive, and the exact treatment moves them by at most 0.27 of the shot noise.

Consequences. The exploration variant at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ is safe: at the calibrated biases the region beyond $k_{\rm neg}$ carries a negligible share of its kept bands, and one prior width below the calibrations every kept C_ℓ stays positive. At the fiducial widths' higher cuts on the dense low-redshift BGS, one-loop perturbation theory is not trustworthy near $k \approx 0.15$–$0.2\,h\,{\rm Mpc}^{-1}$ once b_2 sits a unit below the calibration (Figure 5). A Fisher forecast at the fiducial is unaffected; chains that reach such points meet the open issue of positivity (Appendix D). Two levers miss the sign problem: a tighter b_2 prior only shifts $k_{\rm neg}$, and SN_0 adds the same constant at every k. The subtracted convention is also why SN_0 keeps DESI's width in Poisson units: in the unsubtracted convention SN_0 would have to absorb 5–42 Poisson units in the forecast's BGS, far outside a width of $2/\bar{n}$.

9 Validation

Each piece of the model was checked against an independent calculation, and the forecast built on it was tested against truths it does not contain. The checks of the α_2 projection and of the white-noise splice are in Sections 8.2 and 8.4.

- **Loop templates.** With $b_{3\rm nl} = 0$, CosmoLike's C-FASTPT templates and Python FAST-PT agree to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ on the 62 Mpc vector, cross pairs included ($\Delta\chi^2$: the χ^2 of the difference of the two data vectors under the forecast covariance). Doubling the C-FASTPT grid moves $\Delta\chi^2$ by $\sim 10^{-8}$, raising `accuracyboost` to 3–4 moves the one-loop part by $\Delta\chi^2 \leq 4.5 \times 10^{-3}$, and the low-rank rank is converged at 5 (5 to 12: 10^{-10}). One setting fails: `internal_accuracyboost` = 4 breaks the C-FASTPT bias tables ($\Delta\chi^2 \sim 10^{12}$).

- **Cross pairs.** The one-loop pair code leaves linear-bias vectors bitwise unchanged with the low-rank, FKEM and Limber engines. A pair of two bins with identical redshift distributions equals the polarization of the auto spectra – the bilinear form recovered from the quadratic one, as $u_i^{\mathsf{T}} Q u_j = \frac{1}{4}\left[(u_i+u_j)^{\mathsf{T}} Q (u_i+u_j) - (u_i-u_j)^{\mathsf{T}} Q (u_i-u_j)\right]$ – to 10^{-14} of the one-loop increment. One and eight threads give identical vectors.

- **The likelihood option.** A zero amplitude reproduces the vector without the term bitwise. The α_2 and SN_2 terms are linear (forward and backward differences of the α_2 columns agree to 3×10^{-10} of the column; SN_2 to 3×10^{-14}), and the stochastic-template code run at order zero reproduces the SN_0 templates of the Fisher notebook's derivative caches exactly.

- **The covariance.** Its one-loop increment equals an independent evaluation of the data-vector formula to 1.5×10^{-13} (gg) and 3.2×10^{-12} (gs), and its mean signal divided by the one-loop fiducial data vector is flat at 1.03, the massless-neutrino offset.

- **The derivatives.** The per-bin bias columns of the Fisher construction, summed through the chain rule of Eq. (13), reproduce the stencil derivative of each group amplitude (the correlated-prior note, on its numerical checks).

- **The forecast.** The correlated-prior note fits random bias histories under every linear-bias prior (its accuracy test), and the Fisher notebook of the paper model, `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb`, fits the fiducial model with α_2 free, without

and with tilts, to four fixed truths: α_2 at one prior width ($0.5/k_{\rm max}^2$), $\mathrm{SN}_2 = \pm 5$, the offsets at one sigma of the fiducial widths, and a linear drift of the calibrations across each group (Δ_2 changing by 1 and Δ_s by 0.5, with zero mean). The first three lie inside the fitted prior and are reported; the drift is tested, and a model holds at a cut while every tested shift stays below $\sigma/3$. The fit without tilts on the baseline prior breaks at $0.075\,h\,\mathrm{Mpc}^{-1}$ on the two thin layouts (the drift moves (w_0, w_a) by 0.82 and 0.86σ at 0.1) and at 0.05 on $\Delta z = 0.05$ and 0.2 (0.35 and 0.36σ at 0.075). The fit with tilts on the pessimistic prior absorbs the drift exactly and passes every cut; its reported in-prior shifts reach 0.98σ at 0.075 on $\Delta d_{\rm com} = 62\,\mathrm{Mpc}$ and 1.10σ at 0.1 on $\Delta z = 0.02$, both from α_2. The paper model (Section 7.4) fixes $\alpha_2 = 0$; fitted to a truth with $\alpha_2 = \pm 12.5\,h^{-2}\mathrm{Mpc}^2$ it moves no scored parameter by more than 0.33σ (`notes/fog_parameter_degeneracy_study`). The correlated-prior note's figure `fom_kmax_models` draws the figures of merit of both scenarios and the largest shift against $k_{\rm max}$.

A Bases and conversions, with every factor

A.1 The operators

The tidal field of a density field δ is $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\delta$, with $s^2 = s_{ij}s_{ij}$. Writing the normalized potential $\phi = \partial^{-2}\delta$, expanding the square gives $s^2 = (\partial_i\partial_j\phi)^2 - \frac{2}{3}\,\delta\,\partial^2\phi + \frac{1}{9}\,\delta_{ij}\delta_{ij}\,\delta^2 = (\partial_i\partial_j\phi)^2 - \frac{1}{3}\,\delta^2$, using $\partial^2\phi = \delta$ and $\delta_{ij}\delta_{ij} = 3$. The Galileon operator is $\mathcal{G}_2 = (\partial_i\partial_j\phi)^2 - \delta^2$, so

$$\mathcal{G}_2 = s^2 - \frac{2}{3}\,\delta^2. \tag{23}$$

A term $b_{\mathcal{G}_2}\mathcal{G}_2$ in δ_g therefore equals $b_{\mathcal{G}_2}s^2 - \frac{2}{3}b_{\mathcal{G}_2}\delta^2$: matching to Eq. (1), $b_{s^2} = 2\,b_{\mathcal{G}_2}$, and the quadratic coefficients of the two bases relate as $b_2 = b_2' - \frac{4}{3}\,b_{\mathcal{G}_2}$, where b_2' is the coefficient of $\delta^2/2$ in the $(b_2', b_{\mathcal{G}_2})$ basis. The third-order operator of Eq. (1) is $\psi = \eta - \frac{2}{7}s^2 + \frac{4}{21}\delta^2$ with $\eta = \theta - \delta$ the velocity-divergence difference [1, 4]; at one loop only one combination of the cubic operators survives renormalization, with the filter $\sigma_3^2(k)$ of [2] as its power-spectrum template.

A.2 Normalization bookkeeping

Three conventions circulate for the quadratic and tidal terms, and every difference discussed in Section 5 is a crossing between two of them:

convention	quadratic term	tidal term
CCL / FAST-PT / [2]	$(b_2/2)\,\delta^2$	$(b_{s^2}/2)\,s^2$
velocileptors Eulerian [4]	$(c_2/2)\,\delta^2$	$c_s\,s^2$
velocileptors Lagrangian [4, 5]	$(b_2^{\rm L}/2)\,\delta_0^2$	$b_s^{\rm L}\,s_0^2$

So $b_2 = c_2$ but $b_{s^2} = 2c_s$: the quadratic parameters agree across the Eulerian conventions while the tidal ones differ by the factor of two that Eq. (5) carries.

A.3 The rotation, from the co-evolution solution

The 8/21 and $-2/7$ of Eq. (4) follow from solving the coupled continuity and Euler equations for a tracer that starts with local Lagrangian bias and is then advected (the derivation followed

here is [2]; the same coefficients appear as the bias-parameter rotation in [4]). At linear order the Lagrangian bias dilutes as $b_1^{\rm L}(z) \propto 1/D(z)$, equivalently $b_1(z)-1 \propto 1/D(z)$: the co-evolution statement and the growth-tied shape $B_g/D(z)$ coincide for the linear bias up to the additive 1. At second order the evolved overdensity picks up, next to the gravitational F_2 kernel, the terms

$$\delta_g^{(2)} \supset \left[\frac{b_2^{\rm L}}{2} + \frac{4}{21} b_1^{\rm L}\right] \delta^2 \; - \; \frac{2}{7}\, b_1^{\rm L}\, s^2, \tag{24}$$

which, matched to $(c_2/2)\delta^2 + c_s s^2$, give $c_2 = b_2^{\rm L} + \frac{8}{21} b_1^{\rm L}$ and $c_s = b_s^{\rm L} - \frac{2}{7} b_1^{\rm L}$ once an initial $b_s^{\rm L} s_0^2$ is carried along. At third order the same procedure yields the dynamically generated piece $b_{3\rm nl} = \frac{32}{315}\, b_1^{\rm L}$ of Eq. (6) [2]. In the CCL normalization the tidal entries double, as Eq. (5) records.

A.4 Worked example: LRG1

DESI samples $(1 + b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$, $b_s^{\rm L}\sigma_8^2$ at $z_{\rm eff} = 0.510$. Suppose the chain value of the first is 1.448 and $\sigma_8(0.510) = 0.649$ at the fiducial cosmology. Then $b_1 = 1.448/0.649 = 2.23$, and the calibrated fiducials are $b_{s^2} = -\frac{4}{7}(1.23) = -0.70$, $b_{3\rm nl} = \frac{32}{315}(1.23) = +0.125$ and $b_2(b_1{=}2.23) = +0.34$ from Eq. (7). Using 1.448 itself as an Eulerian bias, as the draft table does, puts b_1 35% low and the leading term $b_1^2 P_{\rm mm}$ of P_{gg} 58% low, before the tidal-sign error enters.

B The loop templates and their low-k constants

With F_2 and $S_2 = \mu_{12}^2 - \frac{1}{3}$ the standard second-order density and tidal kernels (μ_{12} the cosine between the two internal wavevectors), the quadratic templates of Eq. (2) are

$$\begin{aligned}
P_{\delta_1\delta_2}(k) &= 2\int_{\mathbf{q}} F_2(\mathbf{q}, \mathbf{k}-\mathbf{q})\, P_{\rm lin}(q)\, P_{\rm lin}(|\mathbf{k}-\mathbf{q}|), & P_{\delta_2\delta_2}(k) &= 2\int_{\mathbf{q}} P_{\rm lin}\, P_{\rm lin}, \\
P_{\delta_1 s_2}(k) &= 2\int_{\mathbf{q}} F_2\, S_2\, P_{\rm lin}\, P_{\rm lin}, & P_{\delta_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2\, P_{\rm lin}\, P_{\rm lin}, \\
P_{s_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2^2\, P_{\rm lin}\, P_{\rm lin}, & &
\end{aligned} \tag{25}$$

(arguments as in the first line throughout), evaluated by FAST-PT as FFTLog convolutions. The $k \to 0$ limits are $P_{\delta_2\delta_2} \to 2\sigma_4$, $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$, $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$ with $\sigma_4 = \int_{\mathbf{q}} P_{\rm lin}^2(q)$, white-noise constants that renormalize into the stochastic sector; CCL's `sub_lowk` and CosmoLike subtract exactly these three constants in the combinations of Eq. (2). The third-order template is $\sigma_3^2(k) P_{\rm lin}(k)$ with the filter σ_3^2 defined through the I_R kernel of [2]; its co-evolution-weighted contribution is the $b_1 b_{3\rm nl}$ curve of Figure 3. The unequal-time factors of Section 8.1 follow from the perturbative order of each side of the correlator: 22-type products of two second-order fields carry $D^2(z_1)D^2(z_2)$, 13-type products of a third-order and a linear field the symmetrized $D^3 D$.

C Numerical setup for the figures

Every figure of this note lives in its own folder under `plots/`, named after the figure, with its data (`data.json`) and the script that draws it from those data alone; `plots/make_figure_data.py` writes every `data.json`, in the group's `ccl` conda environment (CAMB 1.6.5 and FAST-PT 4.0; `cocoapy311` carries neither).

Figures 1, 2, 3 and 4 use setup A of Section 2 (`SETUP_A` of the generator): $\Omega_m = 0.3$, $\Omega_b = 0.04$, $h = 0.6732$, $10^9 A_s = 2.1$ with A_s the primordial scalar amplitude, $n_s = 0.966$, massless neutrinos, giving $\sigma_8(0) = 0.843$ (the likelihood fiducial, with $n_s = 0.96605$ and one massive neutrino of 0.06 eV, has $\sigma_8(0) = 0.8278$). The linear spectrum and $P_{\rm mm}$ come from CAMB, $P_{\rm mm}$ in the Takahashi halofit the likelihood yamls set (CAMB's default nonlinear model, HMcode-2020, moves the percentages of Figures 3 and 4 by at most 0.19 points). The loop templates come from FAST-PT on a 1200-point logarithmic grid with low- and high-k extrapolation and an FFTLog window of 0.75, CCL's `EulerianPTCalculator` defaults; growth factors from the evolution of the $k = 0.05\,\mathrm{Mpc}^{-1}$ mode. The thin-bin densities of Figure 4 come from the project files `data/lssty1xdesi_desi_dr3_ngal.txt` (surface density per bin) and `..._bins.txt` (bin mean redshifts), converted with $\bar{n} = n_{2D}/(\chi^2 \Delta\chi)$ at $\Delta\chi = 62.28\,\mathrm{Mpc}$. The DESI fiducial differs from this cosmology by a few percent in σ_8, which moves the implied biases of Figure 2 coherently by the same few percent and does not affect the identification.

Figure 5 and the σ_4 of Eq. (17) use setup B instead (`SETUP_B` of the generator), the likelihood's evaluation cosmology: the same parameters with $n_s = 0.96605$ and one massive neutrino of 0.06 eV inside Ω_m, the linear spectrum on a 2000-point logarithmic grid up to $40\,h\,\mathrm{Mpc}^{-1}$, and the growth factor from the ratio of the linear spectra at $k = 10^{-4}\,h\,\mathrm{Mpc}^{-1}$. The massless setup gives a σ_4 larger by 7% and values of $k_{\rm neg}$ within 2% of these.

D Open issues

- **The covariance's one-loop terms are Limber.** The covariance's low-ℓ non-Limber recompute is linear, so in the lowest bands of thin bins its one-loop part is the Limber one, an error of at most 1.3% of C_ℓ.

- **The scale cut does not weigh the counterterm and stochastic content.** The paper notebook computes the UV-fraction cut on the paper model's fiducial (b_2 and b_{s^2} on their calibrations, every other higher-order term zero), and the linear-bias experiments keep a cut computed on the linear fiducial (Section 2). At either fiducial the counterterms and stochastic terms vanish, so neither cut sees them: on $\Delta z = 0.05$ at $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ the UV fraction of the α_0 template is a median 0.076 (maximum 0.14) on the kept bands, above the threshold of 0.05, and about half of each bin's SN_2 kernel comes from $k_\parallel$ above the bin's own cut: part of what the fit learns about α_0 and SN_2 comes from scales the cut is meant to exclude.

- **Positivity far from the fiducial.** The low-rank engine floors $P_{gg} - P_\infty$ where it turns negative. At $\alpha_0 = -88.9\,h^{-2}\mathrm{Mpc}^2$, one exploration-variant width below zero, that happens above $k \approx b_1/\sqrt{|\alpha_0|} \approx 0.14\,h\,\mathrm{Mpc}^{-1}$ on 47% of its grid, and the vector curves in α_0 while FKEM's stays linear; on the dense BGS bins at the fiducial widths' cuts it happens once b_2 sits a unit below the calibration (Section 8.4). The Fisher forecast at the fiducial is unaffected, but a chain that reaches such points leaves the validated model. The remedies, a taper of the loop terms above the scale where one-loop perturbation theory is trusted or a positivity prior on P_{gg}, are not implemented.

- **The $b_{3\rm nl}$ template is wrong.** CosmoLike's C-FASTPT $b_{3\rm nl}$ template (`tab_d1d3` in `pt_cfastpt.h`) is a fixed, cosmology-independent table that disagrees with Python FAST-PT ($\Delta\chi^2 \approx 3.4$ on the 62 Mpc vector with $b_{3\rm nl} = b_1 - 1$), so the model sets $b_{3\rm nl} = 0$.

- **No redshift-space one-loop terms.** CosmoLike has none, and FAST-PT's redshift-space templates are not ported to C: for the μ-dependent loop physics the model relies on the sampled Fingers-of-God σ_v and on α_2.
- **Adjacent cross spectra see the shared edge.** The cross spectra of adjacent thin bins are sensitive to small-scale radial physics at the shared edge, as SN_2 shows (Section 8.3). Both width configurations carry SN_2 for that reason; any other radial physics on that scale reaches the cross spectra the same way.

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